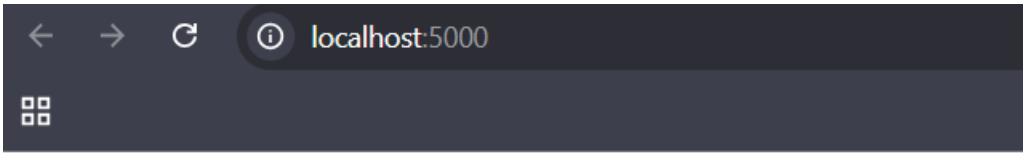
CC LAB 7

Name: Laharish S

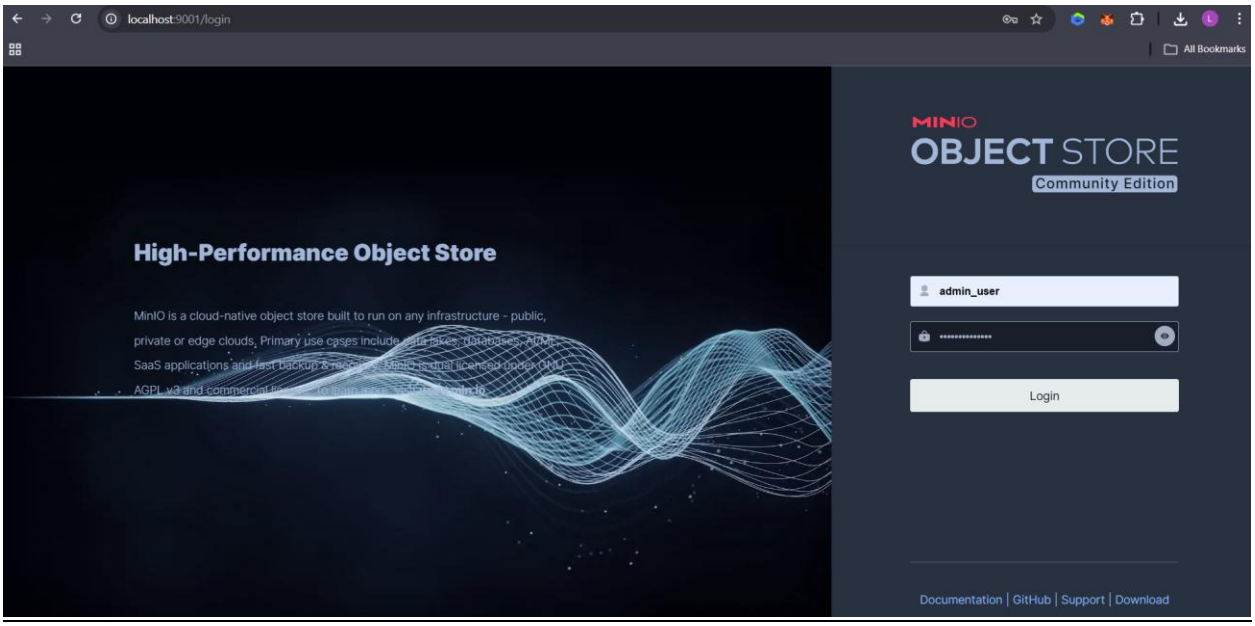
SRN: PES2UG23CS298

SS1

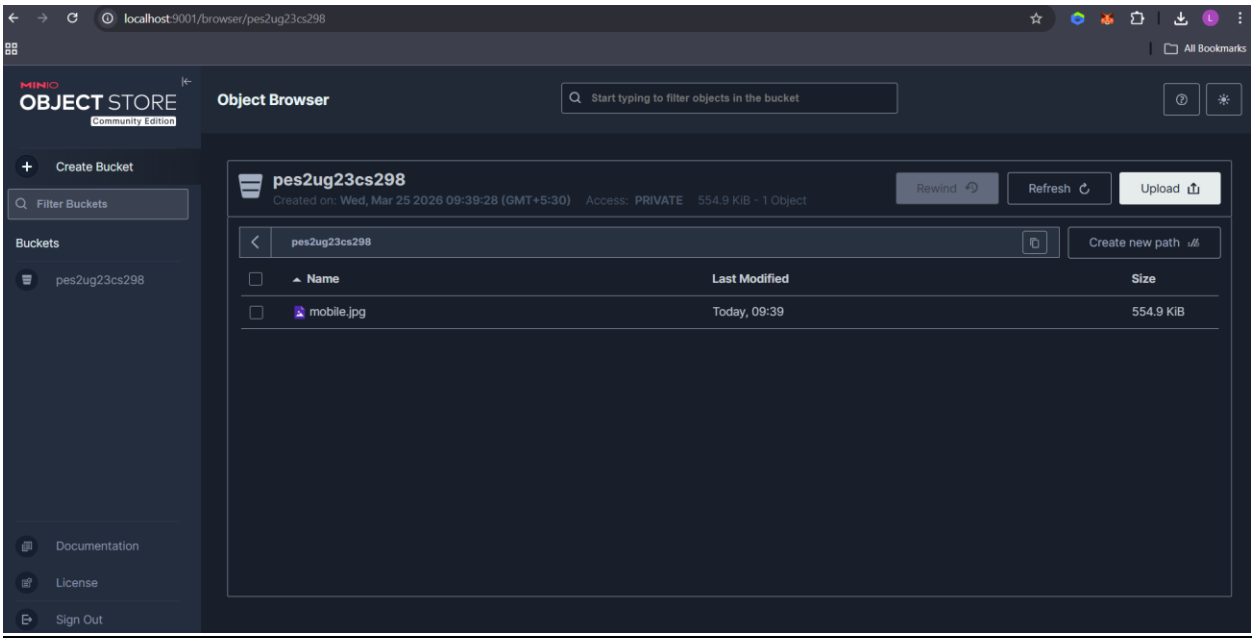


E-commerce Lab is Running! Send a POST request to /product to add data.

SS2



SS3



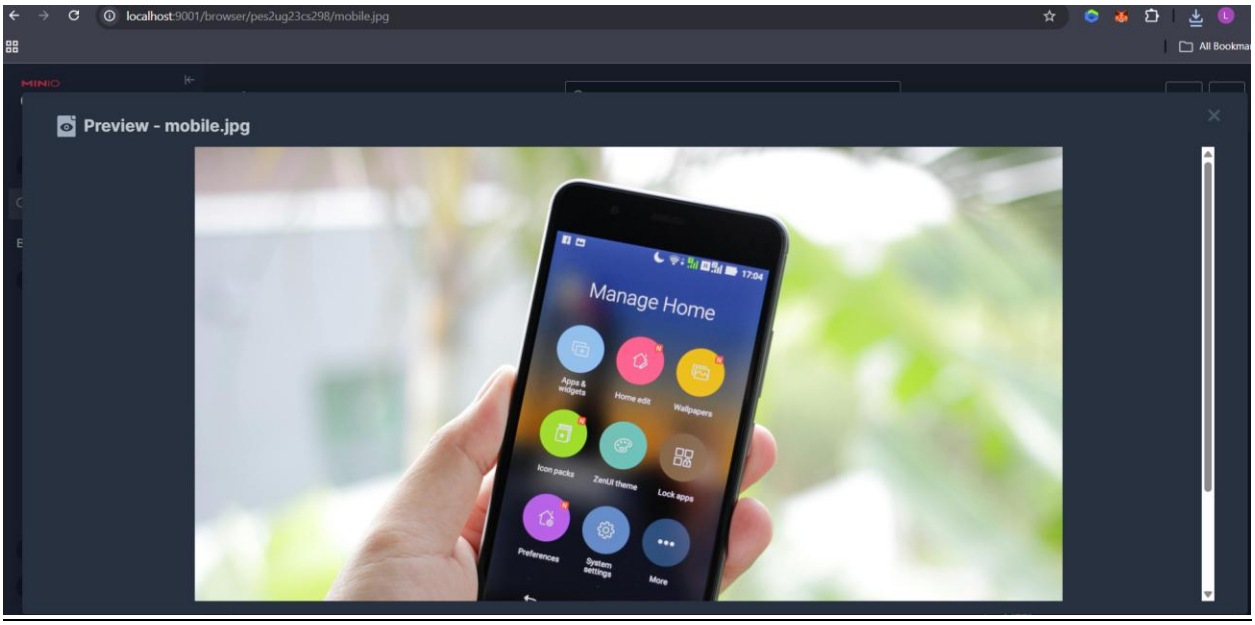

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS D:\LAB\CC\PES2UG23CS298> curl.exe -X POST -F "name=Phone" -F "price=500" -F "image=@mobile.jpg" http://localhost:5000
/product
{"msg":"Structured data in Block, Image in Object","status":"success"}
PS D:\LAB\CC\PES2UG23CS298>
```

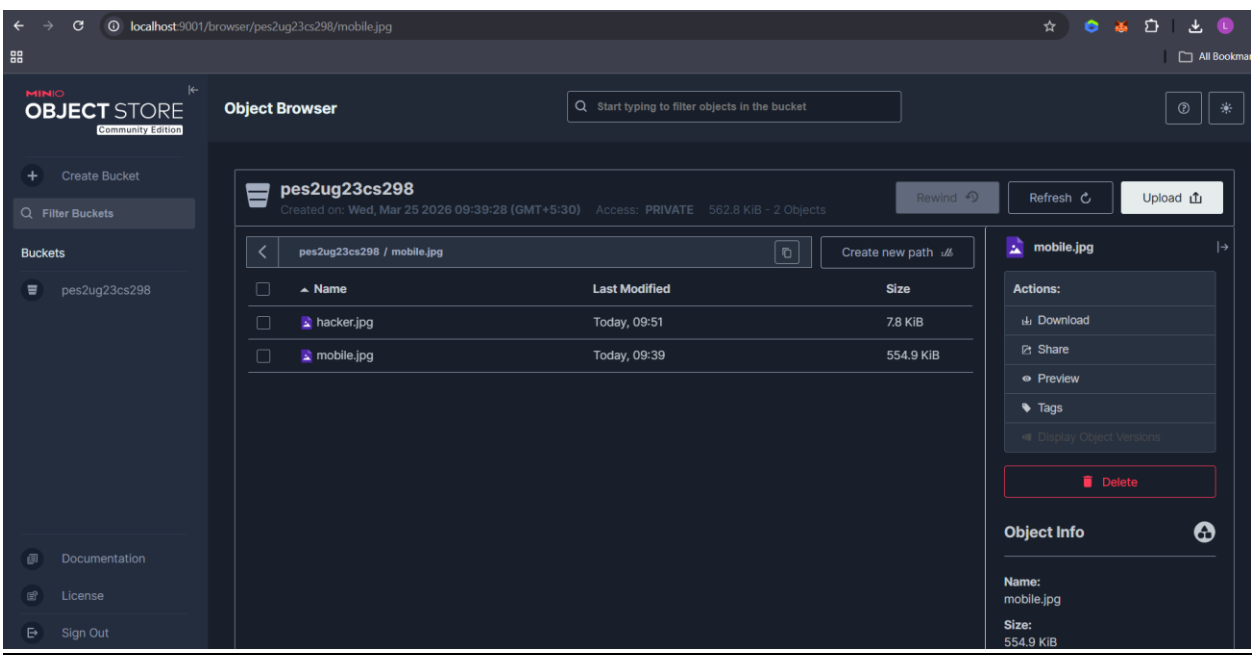
SS4 (Experiment 1)

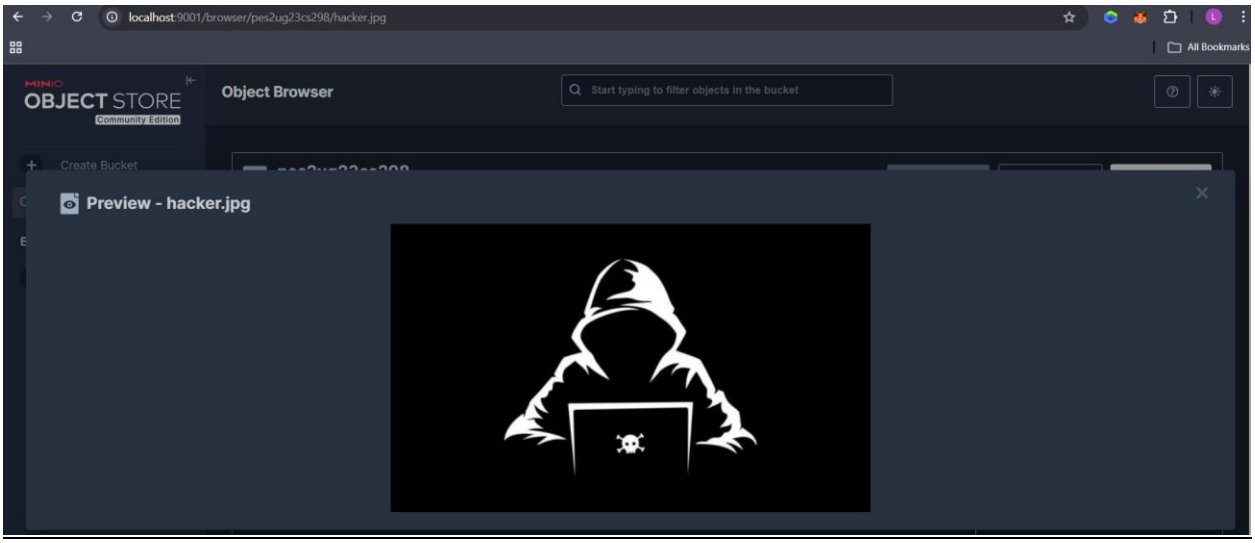
```
PS D:\LAB\CC\PES2UG23CS298> docker exec -it pes2ug23cs298-webapp-1 python -c "import sqlite3; conn = sqlite3.connect('/mnt/block_volume/ecommerce.db'); curs
or = conn.cursor(); cursor.execute('SELECT * FROM product'); print(cursor.fetchall()); conn.close()"
[(1, 'Phone', 500.0, 'mobile.jpg')]
PS D:\LAB\CC\PES2UG23CS298>
```

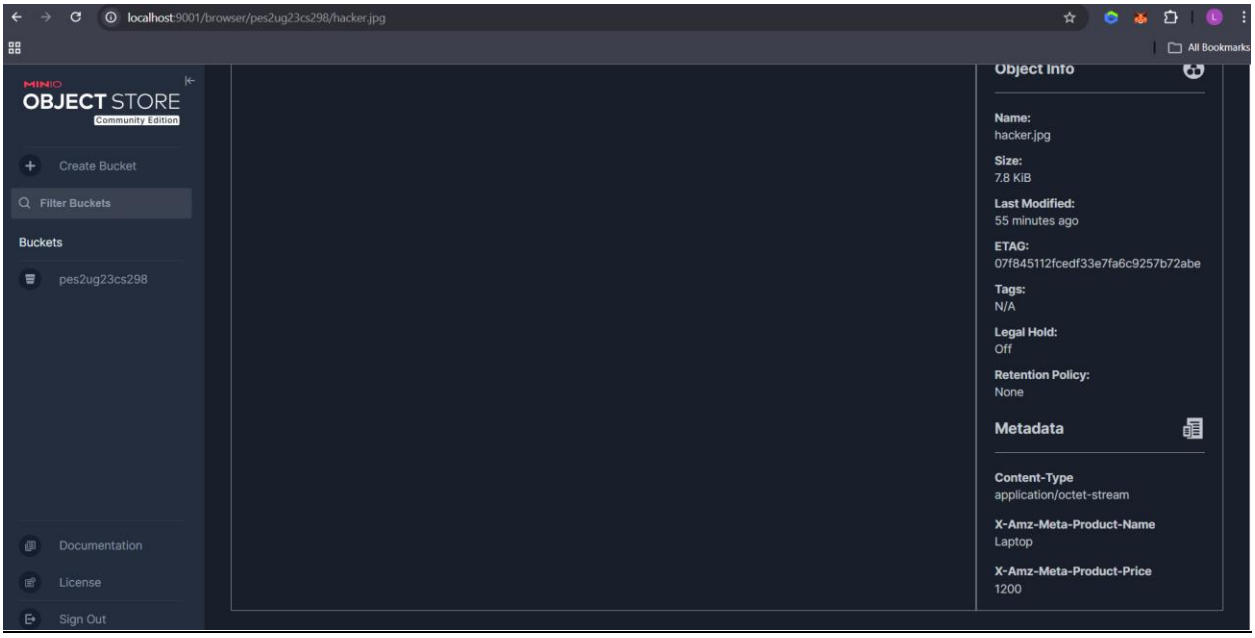


SS5 (Experiment 2)

```
PS D:\LAB\CC\PES2UG23CS298> curl.exe -X POST -F "name=Laptop" -F "price=1200" -F "image=@hacker.jpg" http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":"success"}
PS D:\LAB\CC\PES2UG23CS298>
```




localhost:9001/browser/pes2ug23cs298/hacker.jpg
All Bookmarks
MINIO
OBJECT STORE
Community Edition
Object Browser
Start typing to filter objects in the bucket
Create Bucket
Preview - hacker.jpg


localhost:9001/browser/pes2ug23cs298/hacker.jpg
All Bookmarks
MINIO
OBJECT STORE
Community Edition
Create Bucket
Filter Buckets
Buckets
pes2ug23cs298
Documentation
License
Sign Out
Object Info
Name:
hacker.jpg
Size:
7.8 KiB
Last Modified:
55 minutes ago
ETAG:
07f845112fcedf33e7fa6c9257b72abe
Tags:
N/A
Legal Hold:
Off
Retention Policy:
None
Metadata
Content-Type
application/octet-stream
X-Amz-Meta-Product-Name
Laptop
X-Amz-Meta-Product-Price
1200

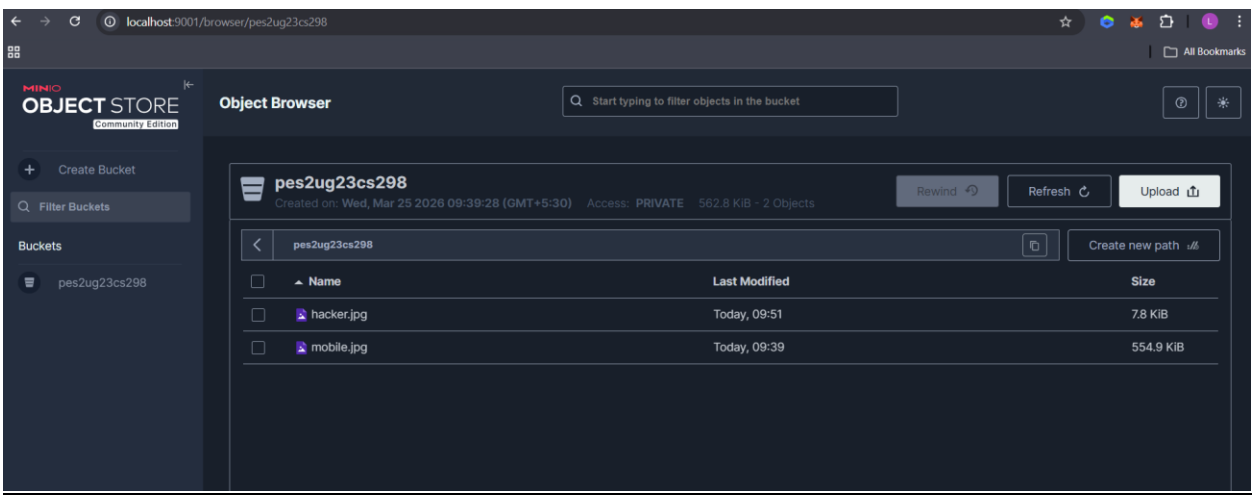
SS6 (Experiment 3)

```
PS D:\LAB\CC\PES2UG23CS298> cd D:\LAB\CC\PES2UG23CS298\my_block_data
PS D:\LAB\CC\PES2UG23CS298\my_block_data> fsutil file createnew dummy.bin 52428800
File D:\LAB\CC\PES2UG23CS298\my_block_data\dummy.bin is created
```

```
PS D:\LAB\CC\PES2UG23CS298> curl.exe -X POST -F "name=Phone" -F "price=500" -F "image=@mobile.jpg" http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":"success"}
PS D:\LAB\CC\PES2UG23CS298>
```

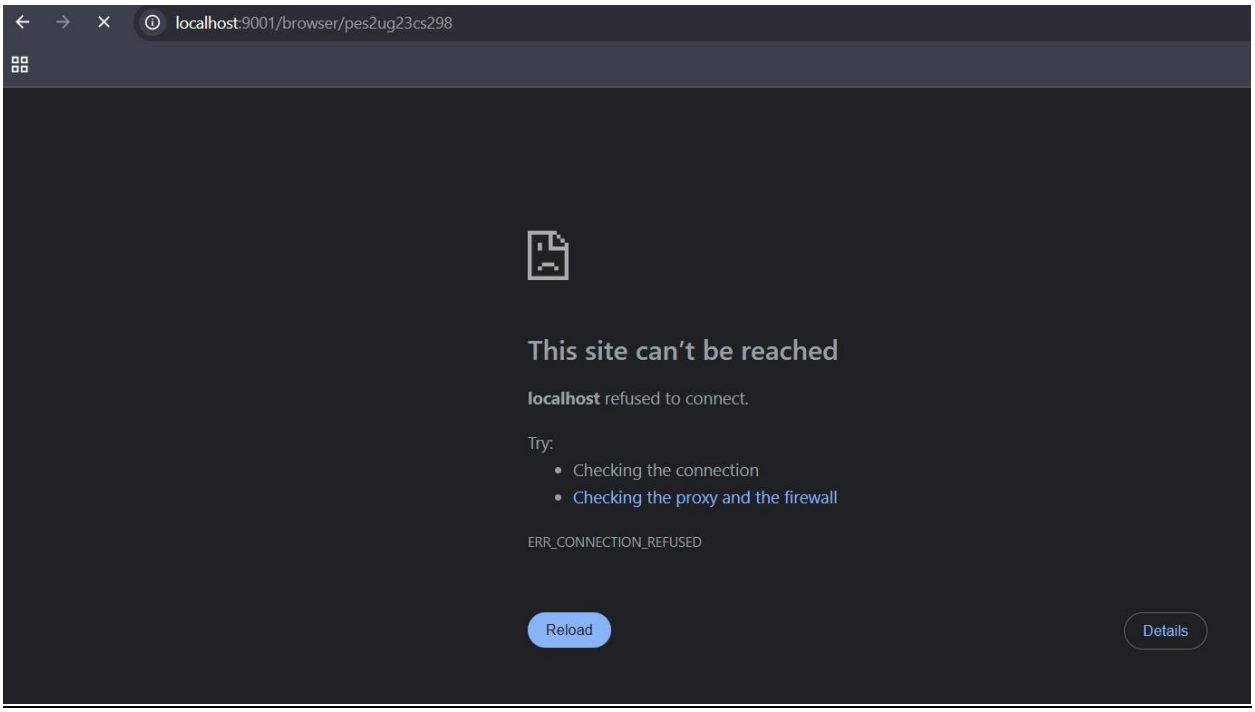
Observation:

Attempted to fill block storage with a 50MB dummy file. The Flask app continued to work as the host disk had sufficient space. However, the concept is demonstrated if the block volume were completely full, Flask would throw a Disk Full error. MinIO continued accepting images independently, showing that Object Storage scales horizontally while Block Storage is constrained by volume size.



SS7 (Experiment 4)

```
PS D:\LAB\CC\PES2UG23CS298> docker-compose stop minio
time="2026-03-25T11:02:22+05:30" level=warning msg="D:\\LAB\\CC\\PES2UG23CS298\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored,
 please remove it to avoid potential confusion"
[+] stop 1/1
 ✔Container pes2ug23cs298-minio-1 Stopped                                                                                                                 0.5s
PS D:\LAB\CC\PES2UG23CS298> curl.exe -X POST -F "name=Phone" -F "price=500" -F "image=@mobile.jpg" http://localhost:5000/product
<!doctype html>
<html lang=en>
<title>500 Internal Server Error</title>
<h1>Internal Server Error</h1>
<p>The server encountered an internal error and was unable to complete your request. Either the server is overloaded or there is an error in the application
.</p>
PS D:\LAB\CC\PES2UG23CS298>
```




```
PS D:\LAB\CC\PES2UG23CS298> docker-compose start minio
time="2026-03-25T11:07:38+05:30" level=warning msg="D:\\LAB\\CC\\PES2UG23CS298\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored,
 please remove it to avoid potential confusion"
[+] start 1/1
 ✔Container pes2ug23cs298-minio-1 Started                                                                                                                 0.4s
PS D:\LAB\CC\PES2UG23CS298> docker-compose stop webapp
time="2026-03-25T11:07:46+05:30" level=warning msg="D:\\LAB\\CC\\PES2UG23CS298\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored,
 please remove it to avoid potential confusion"
[+] stop 1/1
 ✔Container pes2ug23cs298-webapp-1 Stopped                                                                                                                0.0s
PS D:\LAB\CC\PES2UG23CS298> dir D:\LAB\CC\PES2UG23CS298\my_block_data\


    Directory: D:\LAB\CC\PES2UG23CS298\my_block_data


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        25-03-2026     10:59           8192 ecommerce.db


PS D:\LAB\CC\PES2UG23CS298>
```